POE TASK 2

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Agri-Energy Connect Platform- Prototype

Overview:

The Agri-Energy Connect Platform is a prototype digital ecosystem developed to bridge the gap between sustainable agriculture and renewable energy solutions. Built using Visual Studio ASP.NET Core Razor Pages, this platform facilitates knowledge sharing, product procurement, and training for farmers and green energy providers.

Features

1. Home Page: Introduction to the platform and its features, providing a comprehensive overview and easy navigation to other sections.

2. Products Page: View all available products tailored to agricultural needs, complete with detailed descriptions and user reviews.

3. Farmer's Product Management Page: Farmers can view, add, and manage their products, keeping their marketplace presence up to date.

User Role Capabilities:

1. Employees:

-View Products: Employees can browse and review products available in the marketplace.

-Add Farmers: Employees can register new farmers into the platform.

2. Farmers:

-View Products: Farmers can view all available products, focusing on those relevant to their farming needs.

-Add Products: Farmers can list new products under their profile, contributing to the marketplace.

- Manage Own Products: Farmers can view and manage products listed under their name.

How to build and run the prototype

Prerequisites

-Visual Studio 2019 or later

-Git

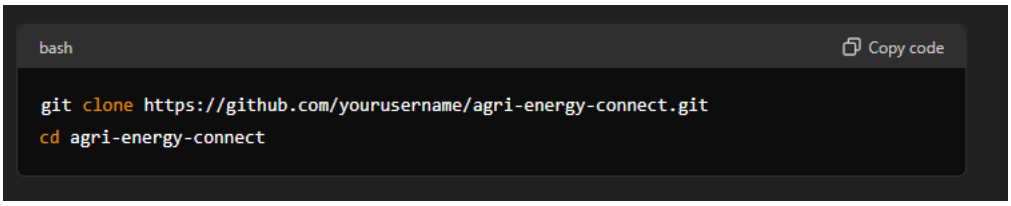
-.NET Core SDK 3.1 or later

-SQL Server LocalDB

Setup Instructions

Step 1: Clone the Repository

-For technical users, clone the repository using Git:



Step 2: Open the Project in Visual Studio

-Launch Visual Studio.

-Open the solution file Agri_Energy_Connect.sln.

Step 3: Restore NuGet Packages

-In Solution Explorer, right-click the solution.

-Select Restore NuGet Packages.

Step 4: Set Up the Database

-Open the appsettings.json file and ensure the connection string is configured for LocalDB:json

-Copy code

"ConnectionStrings": {

"DefaultConnection": "Data Source=(localdb)\\MSSQLLocalDB;Initial Catalog=aspnet-WebApplication1-d4da42dd-ee50-4ff7-aaab-1122c324ceab;Integrated Security=True;Connect Timeout=30;Encrypt=False;Trust Server Certificate=False;Application Intent=ReadWrite;Multi Subnet Failover=False"

}



-Open the Package Manager Console (PMC) in Visual Studio:

-Go to Tools -> NuGet Package Manager -> Package Manager Console.

-Run the following commands to apply the database migrations:

```powershell
Update-Database
```

Step 5: Build the Solution

-In Visual Studio, right-click the solution in Solution Explorer.

-Select Build Solution.

Step 6: Run the Application

-Set AgriEnergyConnect as the startup project.

-Click the Start button or press F5 to run the application.

Step 7: Explore the Platform

-Home Page: Introduction to the platform and its features.

Agri Energy Connect Register Login



SERVICES

Offers best practices, interactive forums, and resources for organic farming, water conservation, and soil health.

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Sustainable Farming Hub

Connects farmers with tailored green energy solutions, such as solar-powered irrigation, wind turbines, and biogas systems.

Green Energy Marketplace

Connects farmers with tailored green energy solutions, such as solar-powered irrigation, wind turbines, and biogas systems.

Educational Resources

Provides online courses, webinars, and workshops on integrating renewable energy technologies into agricultural practices.

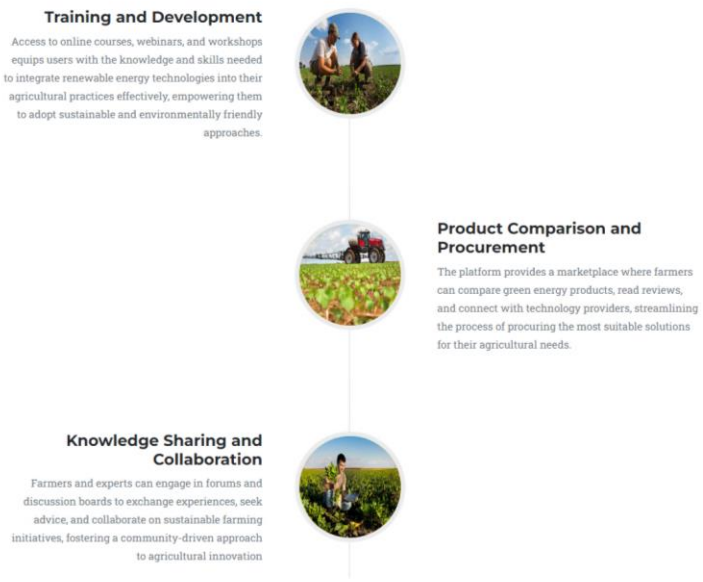
ABOUT

The Agri-Energy Connect Platform bridges sustainable agriculture and renewable energy . Through collaboration, product access, and training, it empowers users to drive positive change in the agricultural sector.

Training and Development

Access to online courses, webinars, and workshops equips users with the knowledge and skills needed

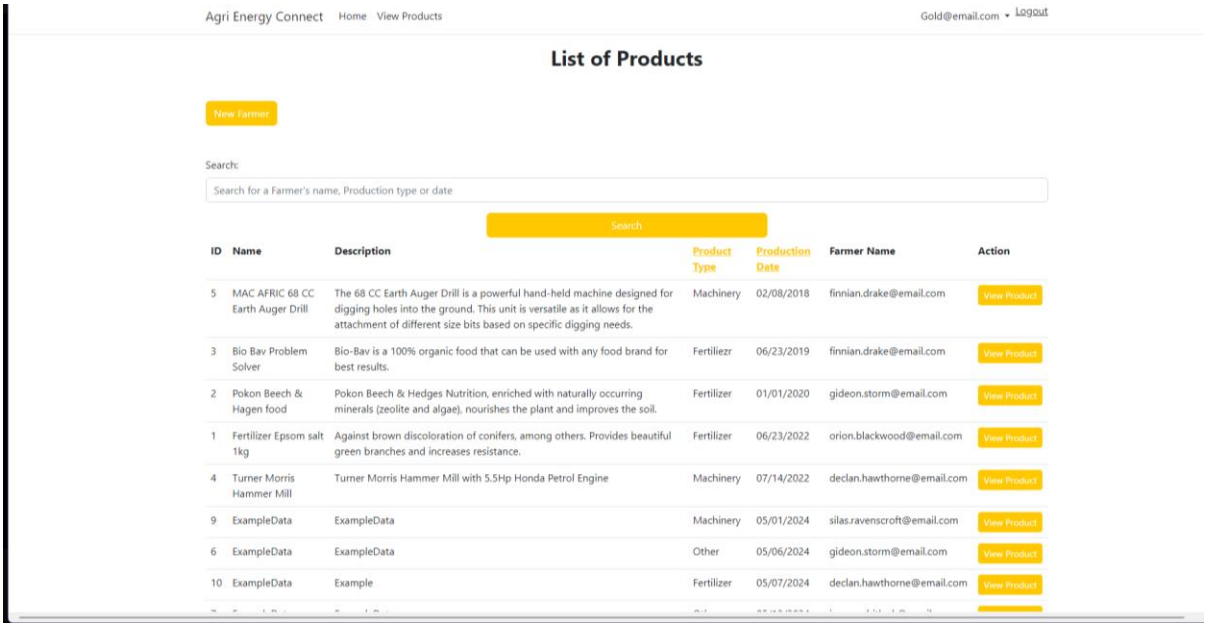




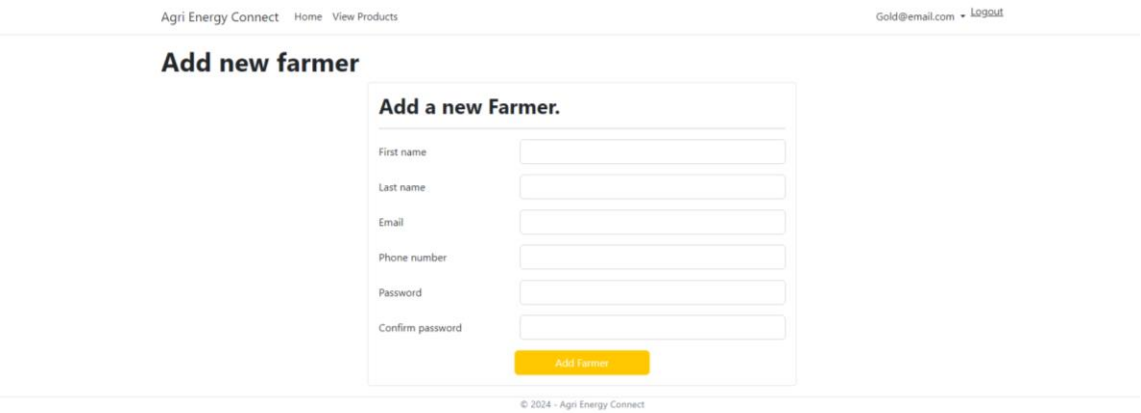
Step 8: User Role Capabilities

1. Employees:

-View Products: Employees can browse and review products available in the marketplace.

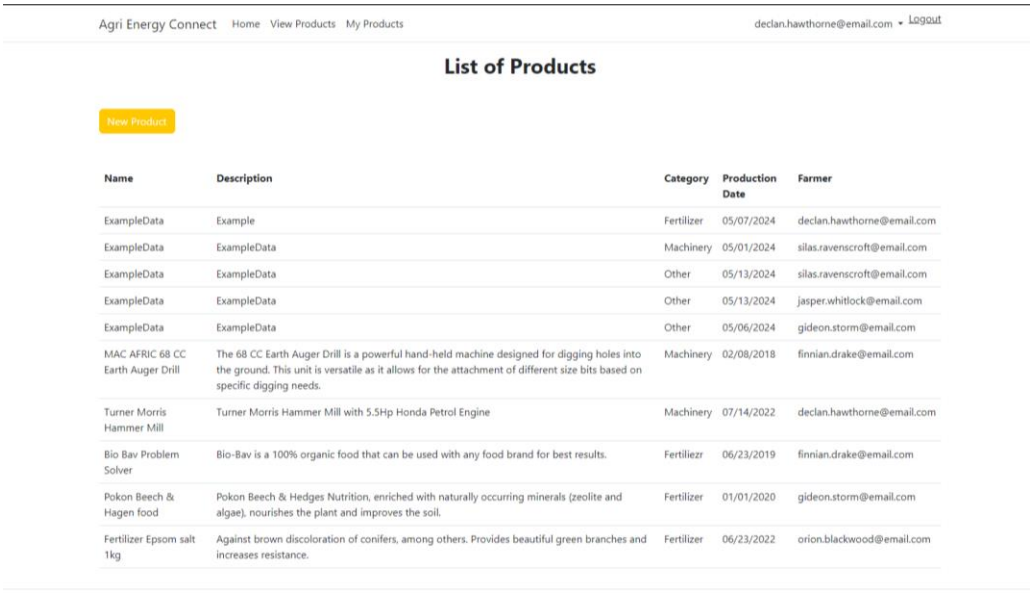


-Add Farmers: Employees have the capability to register new farmers into the platform.



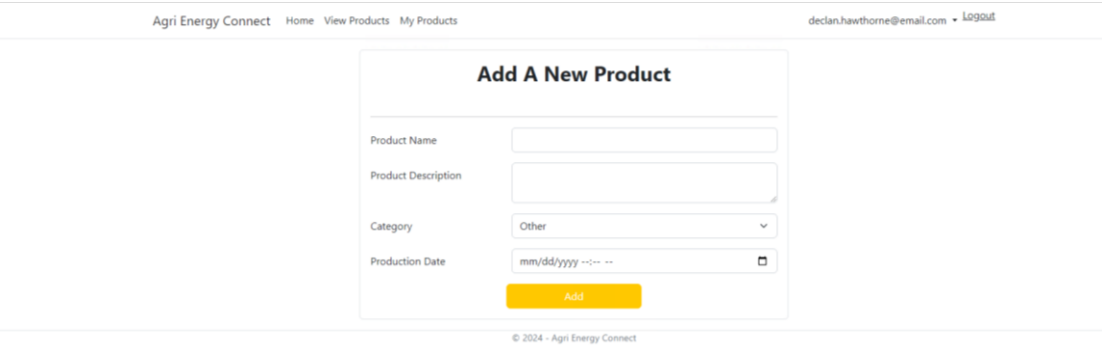
2.Farmers:

-View Products: Farmers can view all available products, focusing on those relevant to their farming needs.

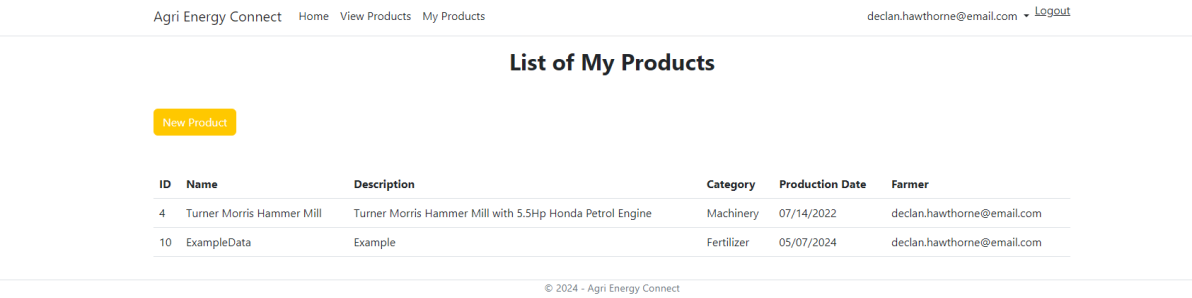


Name	Description	Category	Production Date	Farmer
ExampleData	Example	Fertilizer	05/07/2024	declan.hawthorne@email.com
ExampleData	ExampleData	Machinery	05/01/2024	silas.ravenscroft@email.com
ExampleData	ExampleData	Other	05/13/2024	silas.ravenscroft@email.com
ExampleData	ExampleData	Other	05/13/2024	jasper.whitlock@email.com
ExampleData	ExampleData	Other	05/06/2024	gideon.storm@email.com
MAC AFRIC 68 CC Earth Auger Drill	The 68 CC Earth Auger Drill is a powerful hand-held machine designed for digging holes into the ground. This unit is versatile as it allows for the attachment of different size bits based on specific digging needs.	Machinery	02/08/2018	finnian.drake@email.com
Turner Morris Hammer Mill	Turner Morris Hammer Mill with 5.5Hp Honda Petrol Engine	Machinery	07/14/2022	declan.hawthorne@email.com
Bio Bav Problem Solver	Bio-Bav is a 100% organic food that can be used with any food brand for best results.	Fertiliezr	06/23/2019	finnian.drake@email.com
Pokon Beech & Hagen food	Pokon Beech & Hedges Nutrition, enriched with naturally occurring minerals (zeolite and algae), nourishes the plant and improves the soil.	Fertilizer	01/01/2020	gideon.storm@email.com
Fertilizer Epsom salt 1kg	Against brown discoloration of conifers, among others. Provides beautiful green branches and increases resistance.	Fertilizer	06/23/2022	orion.blackwood@email.com

-Add Products: Farmers can list new products under their profile, contributing to the marketplace.



-Manage Own Products: Farmers can view and manage products listed under their name, ensuring their marketplace presence is up to date.



Troubleshooting

Common Issues

-Database Connection Errors: Ensure SQL Server LocalDB is installed and running. Verify the connection string in appsettings.json.

-NuGet Package Restore Failures: Run dotnet restore from the command line to restore packages manually.

-Build Errors: Check the Error List in Visual Studio for specific issues and resolve them before rebuilding.